ORTSOC

The Nation's First Cybersecurity Teaching Hospital

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| 0 | **Project Title** | Designing and Implementing a Browser Sandbox for Secure Web Browsing | |
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| 1 | **Project Purpose** | Design and build a secure browser sandbox that allows ORTSOC users to safely investigate potentially malicious websites. Accessible through a web client, the system will run browser sessions in an isolated Virtual Machine (VM) environment, preventing any harm to ORTSOC workstations. This will be an upgrade to existing solutions, improving analyst efficiency without pay-model restrictions such as imposed time limits | |

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| 2 | **Project Scope** | | |
| 2.1 | | **In Scope**<br>(Activities / outcomes the project will provide) | **Details** |
| | 2.1.1 | Research existing solutions | Analyze Browserling, browser virtualization, Dockerized browser, and VM-based sandboxes to identify pros/cons. |
| | 2.1.2 | Research hosting solutions | Research options for hosting short-term and light-weight VMs that can be accessed from a browser. Determine if hosting should be done on existing hardware, new hardware, or cloud infrastructure. |

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| | **2.1.3** | Research VM images | Determine the best image to use in the VMs, minimizing resource usage while maximizing performance. |
| | **2.1.4** | Research Hypervisor | Research potential hypervisor solutions to manage the VMs. |
| | **2.1.5** | Webclient design | Design a simple to use website that allows efficient VM session starts and stops. |
| | **2.1.6** | Secure architecture and session handling | Define how the sandbox will isolate web sessions for web browser instances. Ensure the sandbox session resets after each use to prevent malware persistence. |
| | **2.1.7** | Prototype development | Build a functional demo where a user inputs a URL and views a site in a secure environment. |
| | **2.1.8** | Browser options | Determine the feasibility of integrating multiple browser choices. |
| | **2.1.9** | Documentation | Writing documentation for how to use the sandbox. |

| **2.2** | | **Out-of-Scope**<br>(Activities / outcomes the project will NOT provide) | **Details** |
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| | **2.2.1** | Enterprise deployment | Rolling out a fully scaleable production ready to use for thousands of users. |
| | **2.2.2** | Mobile integration | Supporting sandbox browsing on iOS/Android devices. |

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| | **2.2.3** | Full VM use | Webclient will not provide a fully functional Operating System (OS) in the VM. |
| | **2.2.4** | Use for casual browsing | The Web Client will not be intended for everyday web browsing. |
| | **2.2.5** | Cloud infrastructure scaling | Building a cloud infrastructure adaptable for heavy load use. |
| | **2.2.6** | User management system | Handling authentication, account limits, etc. |

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| **2.3** | **Success Criteria** | Metrics for what constitutes success from the project | |
| | **2.3.1** | Develop a functioning sandbox prototype that isolates browsing activity, with the ability to safely load and view potentially malicious sites. | |
| | **2.3.3** | Prototype follows a cost-effective design with fewer usage restrictions than Browserling. | |
| | **2.3.4** | Documentation and future recommendations for ORTSOC's Security Engineering Leads. | |
| | **2.3.5** | Can use webclient to start and stop a VM session. | |
| | **2.3.6** | VM is limited to only provide a browser and no other OS capabilities. | |
| | **2.3.7** | Prototype does not have prohibitive usage limits. | |

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| **3** | **Project Logistics** | Describes who is doing what for the project including each person involved and the role they will play | |
| **3.1** | | **Roles & Responsibilities** | **Role & Name** |
| | **3.1.1** | | Sponsor |
| | **3.1.2** | | Stakeholder |
| | **3.1.3** | Manages the team's performance and project communication | Project Manager<br>Gavi Fifer |
| | **3.1.4** | Webwork, documentation | Project Team Member<br>Carter McQuigg |
| | **3.1.5** | Hypervisor | Project Team Member<br>Tyler Knudson Forrest |
| | **3.1.6** | Front end webpage | Project Team Member<br>Jinhui Zhen |

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| | **3.1.7** | Documentation, backend/connectors | Project Team Member Christian Delisle |
| | **3.1.8** | Front end | Project Team Member Jordan Cowan |

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| **3.2** | **Schedule** | When the projects starts and when it will end. Deeper details are found in the Timeline and Budget tabs | |
| | **3.2.1** | Start Date | 9/3/2025 |
| | **3.2.2** | End Date | 12/9/2025 |

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| **4** | **Requirements** | The activities required to achieve the purpose of the project, as defined by the project Sponsor and Stakeholder(s), listed with detail.<br>Are the dependencies for each requirement defined? | |
| | | **Requirement** | **Dependencies** |
| | **4.1** | Research existing sandbox/browser isolation solutions. | Look at documentation, and consult ORTSOC Security Engineering Leads for guidance. |
| | **4.2** | Design system architecture for a cost-effective browser sandbox. Test prototype with sample suspicious/malicious URLs in a controlled environment. | Decision on whether to use Docker, VMs, or other technology from ORTSOC for environment setup. ORTSOC approved test URLs, sandbox lab environment, and safety guidelines. |
| | **4.3** | Project is approved by sponsors and stakeholders | Project plan is thorough, provides value, and has a clear end goal. |
| | **4.4** | Planning | A plan has been mapped out and major issues considered. |
| | **4.5** | Avoid usage limits | Can be used by multiple analysts at once. |
| | **4.6** | Prototype testing | Has been thoroughly tested for security and ease of use. |

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| | **4.7** | Documentation | Adequate documentation has been created on how to use the webclient and replicate VMs. |

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| **5** | **Deliverables** | All the deliverables for the project listed, along with their including their acceptance criteria. Corresponding deadlines are developed in the timeline tab | |
| | | **Deliverable** | **Acceptance Criteria** |
| | **5.1** | VM tutorial | Provides a basic walkthrough for users with no prior knowledge of VM machines to access websites safely. |
| | **5.2** | VM Documentation | Clear and thorough documentation explaining every aspect and feature of the VM |
| | **5.3** | Project charter with timeline and milestones | Clear and concise charter that all members have approved with a realistic and manageable timeline. |
| | **5.4** | Webpage | Web client has a simple design, is easy to use, and lightweight. Allows users to quickly build and teardown browser VM sessions. |
| | **5.5** | Team project Report 1 | Clear, concise, and factual report of project status. |
| | **5.6** | Team project Report 2 | Clear, concise, and factual report of project status. |
| | **5.7** | Draft Completion Report | Well rounded draft of the overall progress. |
| | **5.8** | Completion Report Presentation | All team members are present. |
| | **5.9** | Completion Report | Clear and concise report. |

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| | **5.10** | Secure Virtual Machine | The Virtual Machine is secure from malware escaping the system and is able to access dangerous internet websites and reset to a previous uncompromised image afterward. |
| | **5.11** | Accessible Website | The webpage is accessible from an internet browser and connects to the Virtual Machine |

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| **Timeline** | What will happen? When will it be done? And by whom? | | | |
| | Includes requirements & dependencies | | | |
| | Provide the start and end dates for the project, the project duration in person-hours. | | | |
| | | **Requirements & Dependencies** | **Start Date** | **End Date** |
| | Develop Plan | Project is approved by the teacher, group members are all in agreement on the project | 10/3/2025 | 10/17/2025 |
| | Completed Plan | Plan fulfills requirements for the class | 10/17/2025 | 10/17/2025 |
| | Propose Plan | Stakeholders agree to meet with the group and listen to the proposal | 10/17/2025 | 10/17/2025 |
| | Acceptance & Project Start | Project was accepted by Stakeholders and the group is given the go-ahead to start on the project | 10/17/2025 | 10/18/2025 |
| | Decide on Isolation method | Group finds the most cost-effective Isolation method that also works for the project | 10/18/2025 | 10/20/2025 |
| | Design the architecture | Group follows the criteria of Stakeholders and drafts a website based on their needs | 10/20/2025 | 10/24/2025 |

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| | Consult with sponsors/stakeholder | The group meets with stakeholders to review the proposed design, confirm requirements, and make necessary revisions before proceeding. | 10/24/2025 | 10/24/2025 |
| | Architecture Documentation | The finalized architecture is documented in detail, outlining the system components, interactions, and deployment plan. | 10/24/2025 | 10/27/2025 |
| | Design Webpage | The Webpage is designed in a user friendly manner. | 10/27/2025 | 11/7/2025 |
| | Isolation Method Implementation | The architecture is implemented and the isolation method is usable in the webpage. | 10/27/2025 | 11/21/2025 |
| | Testing | Ensure that using the webpage will allow for safe searching by users. | 11/24/2025 | 12/1/2025 |
| | Final Documentation of the Project | Compile all documentation of the project into one document. | 12/1/2025 | 12/5/2025 |
| | Tutorial Creation | Create a tutorial for the project that allows for anyone to use it. | 12/1/2025 | 12/5/2025 |
| | Completetion Report | Present details of the overall process and results to the project sponsor and stakeholders. Identify key lessons from the project and identify areas that can be improved. | 12/5/2025 | 12/9/2025 |

| | **Role** | **Hours per Week** | **Number Of Weeks** | **Total** |
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| | Project Team Manager | 9 | 10 | 90 |
| | Project Team Member | 9 | 10 | 90 |
| | Project Team Member | 9 | 10 | 90 |
| | Project Team Member | 9 | 10 | 90 |
| | Project Team Member | 9 | 10 | 90 |
| | Project Team Member | 9 | 10 | 90 |
| | | | | **540 Total Hours** |

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| **Budget** | Planned Budget Sheet | | |
| | Costs & funding sources | | |
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| | | Funding Sources | Costs |
| | Labor | Sponsors | $10,800 |
| | Hosting Costs | Sponsors | $1,500 |
| | Total | | $12,300 |